Elec-S1 Cheat Sheet – Sections 3A to 3C

Kirchhoff's laws · series & parallel · dividers · Millman · Thévenin & Norton · operating point · power matching

3A — Kirchhoff's laws & their consequences

Box 16 – Kirchhoff's laws:

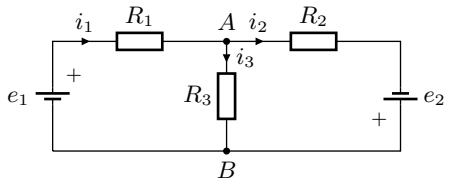
$$\textbf{Node law: } \sum i_{\text{in}} = \sum i_{\text{out}}$$
$$\textbf{Mesh law: } \sum_{\text{mesh}} \varepsilon_k\, u_k = 0$$

- Node law: **charge is conserved** (none piles up at a node).
- Mesh law: **energy is conserved** (round a loop the potential returns to its start). $\varepsilon_k = +1$ if the arrow of u_k points along the direction of travel, -1 otherwise.

Counting: N nodes and B branches give $N-1$ independent node laws and $M = B - N + 1$ independent meshes: B equations for B unknown currents.

Method: put an arrow on each branch current, in any direction (a negative result means it flows the other way); write resistors in receiver convention, $u = Ri$; write the laws, then solve.

Worked example (lecture p. 23):



2 nodes, 3 branches: 1 node law, $3 - 2 + 1 = 2$ meshes.

$$\text{Node } A\text{: } i_1 = i_2 + i_3$$
$$\text{Left: } e_1 = R_1 i_1 + R_3 i_3 \qquad \text{Right: } e_2 = R_2 i_2 - R_3 i_3$$

The outer loop, $e_1 + e_2 = R_1 i_1 + R_2 i_2$, is the sum of the two meshes, so it is not independent.

Consequence 1 – Box 17 – Series (same i; the voltages add) and **Box 18 – Parallel** (same u; the currents add):

$$\text{Series: } R_{\text{eq}} = R_1 + R_2 + \cdots + R_n$$
$$\text{Parallel: } \frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \cdots + \frac{1}{R_n} \qquad \text{two: } R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$$

A series R_{eq} exceeds the largest R_k; a parallel R_{eq} is below the smallest. n identical: nR in series, R/n in parallel. In parallel the conductances add: $G_{\text{eq}} = \sum G_k$.

Consequence 2 – Box 19 – Voltage divider: R_1 and R_2 in series carry the same current $i = u/(R_1 + R_2)$, so

$$u_1 = \frac{R_1}{R_1 + R_2}\, u \qquad u_2 = \frac{R_2}{R_1 + R_2}\, u \qquad \frac{u_1}{u_2} = \frac{R_1}{R_2}$$

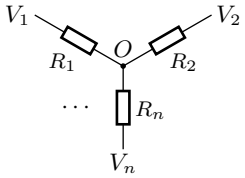
- The larger resistance takes the larger share. $R_1 = R_2$: $u/2$. $R_2 \ll R_1$: $u_2 \approx (R_2/R_1)\, u \approx 0$. $R_2 \gg R_1$: $u_2 \approx u$.
- Valid only if no current leaves the midpoint. A load R_L across R_2 needs $R_L \gg R_2$; otherwise replace R_2 by $R_2 \parallel R_L$ first.
- Uses: reference voltages, matching voltage levels, resistive sensors (thermistors, photoresistors), biasing.

Current divider (two resistors in parallel sharing a current i):

$$i_1 = \frac{R_2}{R_1 + R_2}\, i = \frac{G_1}{G_1 + G_2}\, i \qquad i_2 = \frac{R_1}{R_1 + R_2}\, i$$

Note the crossed indices: the smaller resistance takes the larger current.

Consequence 3 – Box 20 – Millman's theorem (the node law written with potentials). A node O joined only through resistors R_k to points at potentials V_k:



$$V_O = \frac{\dfrac{V_1}{R_1} + \dfrac{V_2}{R_2} + \cdots + \dfrac{V_n}{R_n}}{\dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots + \dfrac{1}{R_n}}$$

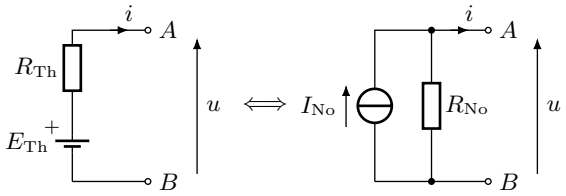
- Proof: node law at O, $\sum_k (V_k - V_O)/R_k = 0$.
- V_O is the mean of the V_k weighted by the conductances $1/R_k$, so it lies between the smallest and the largest V_k.
- A branch to ground puts $0/R_k$ on top, but its $1/R_k$ **stays** in the denominator.
- A branch "source E_k + R_k" to ground: $V_k = +E_k$ if the + terminal is on the node side, $-E_k$ if it is on the ground side. Several resistors in one branch: use their sum.

Example (worked example above, $V_B = 0$; the top of e_2 is its − terminal):

$$V_A = \frac{e_1/R_1 - e_2/R_2 + 0/R_3}{1/R_1 + 1/R_2 + 1/R_3}, \qquad \text{then } i_3 = \frac{V_A}{R_3}$$

3B — Thévenin's & Norton's equivalence theorems

A "black box" made *only* of ideal sources and ohmic conductors (no lamp, motor or diode), seen from its terminals A and B, reduces to **two parameters**:



$u = E_{\text{Th}} - R_{\text{Th}}\, i$ $\qquad$ $i = I_{\text{No}} - u/R_{\text{No}}$

Generator convention: i leaves by A and $u = V_A - V_B$.

Box 21 – Equivalence of the models: they give the same $u(i)$ line, so

$$R_{\text{Th}} = R_{\text{No}} \equiv R \qquad E_{\text{Th}} = R\, I_{\text{No}} \qquad I_{\text{No}} = \frac{E_{\text{Th}}}{R}$$

E_{Th} is the open-circuit voltage ($i = 0$) and I_{No} the short-circuit current ($u = 0$). The arrow of I_{No} points to A, the terminal on the + side of E_{Th}.

Box 22 – Thévenin's theorem: between A and B the network is equivalent to an ideal voltage source E_{Th} in **series** with a resistor R_{Th}.

1. **Sources off** → R_{Th}: the resistance seen between A and B (series and parallel combinations).
2. A–B **open** → E_{Th}: sources on, $i = 0$, and $E_{\text{Th}} = u_{AB}$.

Box 23 – Norton's theorem: the same network is equivalent to an ideal current source I_{No} in **parallel** with R_{No} (also written I_{N}, R_{N}).

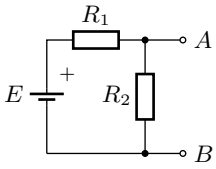
1. **Sources off** → R_{No}: the same study, so $R_{\text{No}} = R_{\text{Th}}$.
2. A–B **short-circuited** → I_{No}: sources on, $u = 0$, and I_{No} is the current in the short circuit.

Switching a source off: an ideal voltage source becomes a **wire** ($u = 0$); an ideal current source becomes an **open circuit** ($i = 0$). The resistors stay in place.

Tools for E_{Th} and I_{No}: voltage and current dividers, Millman, Kirchhoff, and

- **source transformation**: E in series with $R \Leftrightarrow E/R$ in parallel with R (Thévenin models in series add their E and R; Norton models in parallel add their I and G);
- **superposition** (linear circuits): add the effects of each source acting alone, with the other sources switched off.

Worked example (lecture p. 33) – Thévenin model of a voltage divider:

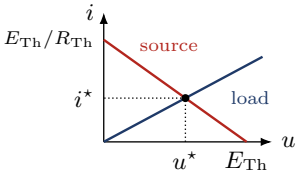


1. E off (a wire): $R_1 \parallel R_2$, so $R_{\text{Th}} = R_1 R_2/(R_1 + R_2)$.
2. Open circuit (divider): $E_{\text{Th}} = R_2 E/(R_1 + R_2)$.
3. $u = E_{\text{Th}} - R_{\text{Th}}\, i$. Check: $I_{\text{No}} = E_{\text{Th}}/R_{\text{Th}} = E/R_1$, the current in R_1 when A–B is shorted. ✓

With $E = 10\,\text{V}$, $R_1 = 2.0\,\text{k}\Omega$, $R_2 = 8.0\,\text{k}\Omega$: $R_{\text{Th}} = 1.6\,\text{k}\Omega$, $E_{\text{Th}} = 8.0\,\text{V}$, $I_{\text{No}} = 5.0\,\text{mA}$.

3C — Operating point of a circuit

A linear network feeding a load R_c (heater, lamp, motor, ...) is first replaced by its Thévenin (or Norton) model.



Source (generator convention): $u = E_{\text{Th}} - R_{\text{Th}}\, i$, a line of slope $-1/R_{\text{Th}}$ in the (u, i) plane.

Load (receiver convention): $u = R_c\, i$, a line through O of slope $1/R_c$.

The **operating point** is the actual state of the circuit: the unique pair $(u^\star, i^\star)$ that satisfies *both* characteristics. Graphically, it is their **intersection**.

Box 24 – Analytical determination: $R_c\, i^\star = E_{\text{Th}} - R_{\text{Th}}\, i^\star$, so

$$i^\star = \frac{E_{\text{Th}}}{R_{\text{Th}} + R_c} \qquad u^\star = R_c\, i^\star = \frac{R_c}{R_{\text{Th}} + R_c}\, E_{\text{Th}}$$

So $u^\star$ is a voltage divider between R_{Th} and R_c.

- $R_c \to 0$ (short circuit): $u^\star \to 0$ and $i^\star \to E_{\text{Th}}/R_{\text{Th}}$; the point reaches the i axis.
- $R_c \to \infty$ (open circuit): $i^\star \to 0$ and $u^\star \to E_{\text{Th}}$; the point reaches the u axis.
- In both cases $P_c = u^\star i^\star = 0$, so the load power has a maximum in between.

Power balance: the load receives exactly what the real source delivers, $P_c = u^\star i^\star = (E_{\text{Th}} - R_{\text{Th}} i^\star)\, i^\star = R_c\, i^{\star 2}$, and

$$\underbrace{E_{\text{Th}}\, i^\star}_{\text{produced by ideal source}} = \underbrace{R_{\text{Th}}\, i^{\star 2}}_{\text{lost in } R_{\text{Th}}} + \underbrace{R_c\, i^{\star 2}}_{\text{received by load}}$$

Efficiency:

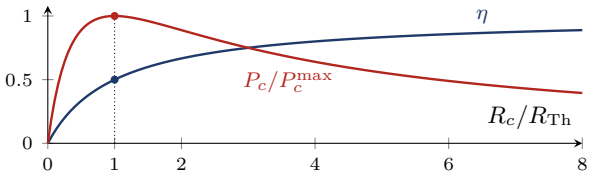
$$\eta = \frac{P_c}{E_{\text{Th}}\, i^\star} = \frac{R_c}{R_{\text{Th}} + R_c} = \frac{u^\star}{E_{\text{Th}}}$$

It increases with R_c: 0 on short circuit, close to 1 for $R_c \gg R_{\text{Th}}$, and exactly $\frac{1}{2}$ at $R_c = R_{\text{Th}}$.

Maximum power – matching:

$$P_c = \frac{R_c\, E_{\text{Th}}^2}{(R_{\text{Th}} + R_c)^2} \qquad \frac{\mathrm{d}P_c}{\mathrm{d}R_c} = E_{\text{Th}}^2\, \frac{R_{\text{Th}} - R_c}{(R_{\text{Th}} + R_c)^3}$$

$$R_c = R_{\text{Th}} \implies P_c^{\max} = \frac{E_{\text{Th}}^2}{4R_{\text{Th}}}, \quad \eta = 50\,\%$$



Maximum power ≠ maximum efficiency, and no load gives both. Match ($R_c = R_{\text{Th}}$) when the available power is tiny (antenna, sensor, amplifier stage). Favour efficiency ($R_c \gg R_{\text{Th}}$) when energy is costly and losses cause heating (power lines, power supplies, a battery driving a motor).

Example: $E_{\text{Th}} = 10\,\text{V}$, $R_{\text{Th}} = 2.0\,\Omega$, $R_c = 3.0\,\Omega$ give $i^\star = 2.0\,\text{A}$, $u^\star = 6.0\,\text{V}$ and $P_c = 12\,\text{W}$. The ideal source produces 20 W and 8.0 W is lost in R_{Th}, so $\eta = 60\,\%$. With a matched load ($R_c = 2.0\,\Omega$): $P_c^{\max} = 12.5\,\text{W}$, but $\eta = 50\,\%$.

Which tool? & exam traps

- One unknown potential (a node joined by resistors) → **Millman**.
- Two resistors in series sharing a voltage → **voltage divider**; in parallel sharing a current → **current divider**.
- A load studied while the rest stays fixed → **Thévenin** or **Norton**, then the **operating point**.
- Anything else → **Kirchhoff**: $N - 1$ node laws and $B - N + 1$ mesh laws.

- A divider needs the **same current** in both resistors: with a load, combine $R_2 \parallel R_L$ first.
- Millman: a branch to ground keeps its $1/R_k$ in the denominator; a reversed source counts as $-E_k$.
- Sources off: voltage source → wire, current source → open. Never remove a resistor.
- In open circuit, a resistor in series with terminal A carries no current: it changes R_{Th}, not E_{Th}.
- $u = E_{\text{Th}} - R_{\text{Th}}\, i$ assumes i leaves by A (generator convention). State your conventions.
- $\eta = P_c/(E_{\text{Th}} i^\star)$, not $P_c/P_c^{\max}$.
- "State a law or theorem": give the words, the diagram or equation, and the method for each parameter (that is how the marks are split).
- Sanity checks: series $R_{\text{eq}} >$ largest R; parallel $R_{\text{eq}} <$ smallest R; V_O lies between the V_k; $0 \le \eta < 1$.

Units: $\text{k}\Omega \times \text{mA} = \text{V}$; $\text{V}/\text{k}\Omega = \text{mA}$; $\text{V}^2/\Omega = \text{W}$; $\text{S} = \Omega^{-1}$.

Handout notes: p. 27, eqs. (15)–(16): the left-hand side is u_2 (the voltage across R_2), not u_1, and "0.9901 E" means 0.9901 u. p. 28, eq. (17): the denominator is $1/R_1 + 1/R_2 + \cdots + 1/R_n$. p. 37, eq. (24): P_{Th} there means $E_{\text{Th}} i^\star$ (produced by the ideal source), not the delivered power of eq. (19).